JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON MONDAY 06th APRIL 2026) **TIME : 3:00 PM TO 6:00 PM**

CHEMISTRY — TEST PAPER WITH SOLUTION

SECTION-A

51. Which of the following contain the same number of atoms ?

(Given : Molar mass in g mol^{-1} of H, He, O and S are 1,4,16 and 32 respectively)

A. 2g of O_2 gas

B. 4g fo SO_2 gas

C. 1400 mL of O_2 at STP

D. 0.05 L of He at STP

E, 0.0625 mol of H_2 gas

Choose the **correct** answer from the options given below :

(1) A and B only (2) B and C only

(3) C and D only (4) A, C and E only

Sol.

$$N_{atoms} \text{ in } O_2 = \frac{m_{O_2}}{M_{O_2}} N_A = \frac{2}{32} \times 2N_A = \frac{N_A}{8} = 0.125\, N_A$$

$$N_{atoms} \text{ in } SO_2 = \frac{m_{SO_2}}{M_{SO_2}} = 3N_A = \frac{4}{64} \times 3N_A = \frac{3N_A}{16} = 0.1875\, N_A$$

$$N_{atoms} \text{ in } O_2 \text{ at STP} = \frac{V_{O_2}}{22.7\,lit} N_A = \frac{1400\,ml}{22400\,ml} \times 2N_A = 0.125\, N_A$$

$$N_{atoms} \text{ in He(g) at STP} = \frac{V_{He}}{22.4\ lit.} N_A = \frac{0.05}{22.4} N_A = 0.002\, N_A$$

$$N_{atoms} \text{ in } H_2(g) = \text{moles} \times \text{atomicity} \times N_A = 0.0625 \times 2N_A = 0.125\, N_A$$

52. The Bohr radius of a hydrogen like species is 70.53 pm. The species and the stationary state(n) are respectively

(Given: Hydrogen atom Bohr radius is 52.9 pm)

(1) Li^{2+}, 3 (2) He^{+}, 3

(3) He^{+}, 2 (4) Li^{2+}, 2

Sol. $r_{species} = a_0 \frac{n^2}{Z}$

$$70.53 = 52.9\left(\frac{n^2}{Z}\right)$$

$$\frac{n^2}{Z} = \frac{70.53}{52.9} = 1.33 = \frac{4}{3}$$

Data will satisfy if n = 2 and z = 3

Li^{+2}, 2 $\Rightarrow \frac{4}{3} = 1.33$

53. Given below are two statements:

Statement I : The number of compounds among SO_2, SO_3, SF_4, SF_6 and H_2S in which sulphur does not obey the Octet rule is 3.

Statement II : Among [H_2O, ClF_3, SF_4], [NH_3, BrF_5, SF_4], [BrF_5, ClF_3, XeF_4] and [XeF_4, ClF_3,H_2O], the number of sets in which all the molecules have one lone pair of electrons on the central atom is 1.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true **Statement II** is false

(4) **Statement I** is false **Statement II** is true

Sol. SO_2 , SO_3 , SF_6 & SF_4 do not follow octet rule

NH_3 , BrF_5 , SF_4 have one lone pair on central atom

ClF_3 , XeF_4 & H_2O have two lone pair on central atom.

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54. Match **List-I** with **List-II**

Given V_1 and V_2 are initial and final volumes respectively:

List-I (Isothermal process)		List-II (Expression)	
A.	Reversible expansion	I.	$q=0$
B.	Free expansion	II.	$q = nRT \ln \frac{V_2}{V_1}$
C.	Irreversible Compression	III.	$w = -p_{ext}(V_1 - V_2)$
D.	Cyclic reversible	IV.	$\frac{q_{rev}}{T} = 0$

Choose the correct answer from the options given below :

(1) A-II, B-III, C-I, D-IV (2) A-II, B-I, C-IV, D-III

(3) A-II, B-I, C-III, D-IV (4) A-I, B-II, C-III, D-IV

Sol. (A) For isothermal process, $\Delta U = 0$

$$W = -nRT\ell n\left(\frac{V_2}{V_1}\right) \text{ so } q = nRT\ell n\left(\frac{V_2}{V_1}\right)$$

(B) Free expansion which is isothermal must be adiabatic, so $q = 0$

(C) Irreversible process $W = -P_{ext}[V_2 - V_1]$

(D) Cyclic process, change in state function (S) so change in entropy must be zero.

$$dS = \frac{dq_{rev}}{T}$$

$$\oint dS = 0$$

55. Given below are two statements :

Chamber 1	Semi-permeable membrane	Chamber 2
18 g glucose in 100 mL aqueous solution		30 g glucose in 250 mL aqueous solution

Statement I : H_2O molecules move from the chamber 1 to chamber 2.

Statement II : The osmotic pressure of a solution prepared by dissolving 50 mg of potassium sulphate (molar mass = 174 g/mol) in 2 L of water (at 27°C) is 0.0107 bar. (Given : R = 0.083dm^3 bar K^{-1} mol^{-1} and assume complete dissociation of electrolyte)

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true but **Statement II** is false

(4) **Statement I** is false but **Statement II** is true

Sol. On side (I) $C_1 = \frac{18/180}{0.1} = 1M$

On side (II) $C_2 = \frac{30/180}{250/1000} = \frac{2}{3}M$; $\{\pi_1 > \pi_2\}$

Hence water will flow from chamber-II to Chamber-I.

$\pi = iCRT$; $i = 3$ for K_2SO_4

And $\left(\pi = 3 \times \frac{50 \times 10^{-3}/174}{2}\right) \times 0.083 \times 300$

= 0.0107 bar.

56. Given is a concentrated solution of a weak electrolyte A_xB_y of concentration 'c' and dissociation constant 'K'. the degree of dissociation is given by:

(1) $[K \times c^{x+y-1}x^x y^y]^{x+y}$

(2) $\left(\frac{K}{c^{x+y-1}x^x y^y}\right)^{\frac{1}{x+y}}$

(3) $\left(\frac{c^{x+y-1}x^x y^y}{K}\right)^{x+y}$

(4) $\left(\frac{c^{x+y-1}x^x y^y}{K}\right)^{\frac{1}{x+y}}$

Sol. $A_xB_y \rightleftharpoons xA^{y+} + yB^{x-}$

$C(1-\alpha)$ $\quad xC\alpha \quad yC\alpha$

$$K = \frac{(xC\alpha)^x (xC\alpha)^y}{C(1-\alpha)}$$

As $\alpha << 1 \Rightarrow (1-\alpha) \simeq 1$

$K = x^x y^y C^{x+y-1} \alpha^{x+y}$

$$\alpha = \left(\frac{K}{C^{x+y-1} x^x y^y}\right)^{\frac{1}{x+y}}$$

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57. For a general redox reaction

Anode $\mathrm{Red}_1 \rightarrow \mathrm{Ox}_1^{n_1^+} + n_1 e^-$

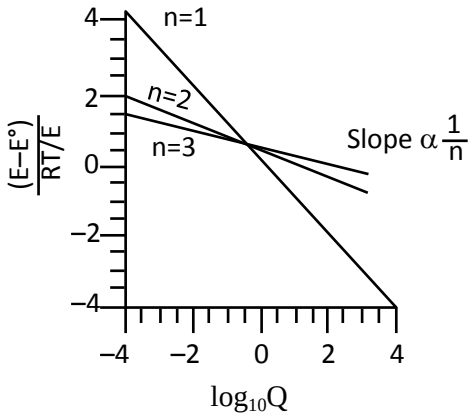
Cathode $\mathrm{Ox}_2 + n_2 e^- \rightarrow \mathrm{Red}_2^{n_2^-}$

Which of the following statement is **incorrect** ?

(1) The overall reaction can be written as

$$n_2\,\mathrm{Red}_1 + n_1\mathrm{Ox}_2 \rightleftharpoons n_2\mathrm{Ox}_1^{n_1^+} + n_1\,\mathrm{Red}_2^{n_2^-}$$

(2) The electrons do not appear in the overall reaction because electrons produced at the anode are consumed at the cathode.



(3) Here n is the number of electrons transferred in redox reaction.

(4) If the reaction is carried out reversibly, the electrical work done is equal to the ratio of charge and potential difference through which charge is moved.

Sol. **Anode :** $\left[\mathrm{Red}_1 \rightarrow \mathrm{Ox}_1^{n_1^+} + n_1 e^-\right] \times n_2$

Cathode : $\left[\mathrm{Ox}_2 + n_2 e^- \rightarrow \mathrm{Red}_2^{n_2^-}\right] \times n_1$

Overall reaction :

$$n_2\,\mathrm{Red}_1 + n_1\mathrm{Ox}_2 \rightarrow n_2\mathrm{Ox}_1^{n_1^+} + n_1\,\mathrm{Red}_2^{n_2^-}$$

The electrons does not appear in overall reaction

$$E = E^\circ - 2.303\,\frac{RT}{nF}\log Q$$

$$\frac{(E - E^\circ)}{RT/F} = -\frac{2.303}{n}\log Q$$

$$\text{Slope} = \frac{-2.303}{n}$$

Straight line passing through origin.

Electrical work = Charge × Potential difference

58. In a period, the first ionisation enthalpy of the element at extreme left and the negative electron gain enthalpy of the extreme right element, except noble gases, are respectively.

(1) lowest and lowest (2) highest and lowest

(3) lowest and highest (4) highest and highest

Sol. Left most element in a period will be of group 1 which will have lowest IE, while right most element will be of group 17 which will have most negative ΔH_{eg}.

59. Given below are two statements :

Statement I : $F_2O < H_2O < Cl_2O$ is the correct trend in terms of bond angle.

Statement II : SiF_4, SnF_4 and PbF_4 are ionic in nature.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true but **Statement II** is false

(4) **Statement I** is false but **Statement II** is true

Sol.

Specie	Bond angle
OCl_2	111°
H_2O	104.5^o
OF_2	103°

SiF_4 is covalent.

60. The correct order of first (Δ_iH_1) and second (Δ_iH_2) ionisation enthalpy values of cr and Mn are :

A. $\Delta_iH_1 : Cr > Mn$

B. $\Delta_iH_2 : Cr > Mn$

C. $\Delta_iH_1 : Mn > Cr$

D. $\Delta_iH_2 : Mn > Cr$

Choose the **correct** answer from the options given below :

(1) A and B only (2) B and C only

(3) A and D only (4) C and D only

Sol. $Cr - [Ar]3d^5 4s^1$

$Mn - [Ar]3d^5 4s^2$

$IE_1 - Mn > Cr$

$IE_2 - Mn < Cr$

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61. Which of the following sequence of hybridisation, geometry and magnetic nature are correct for the given coordination compounds ?

A. $[NiCl_4]^{2-}$ – sp^3, tetrahedral, paramagnetic

B. $[Ni(NH_3)_6]^{2+}$ – sp^3d^2, octahedral, paramagnetic

C. $[Ni(CO)_4]$ – sp^3, tetrahedral, paramagnetic

D. $[Ni(CN)_4]^{2-}$ – dsp^2, square planar diamagnetic

Choose the **correct** answer from the options given below :

(1) A, B, C and D (2) B, C and D only

(3) A, C and D only (4) A, B and D only

Sol. $[Ni(CO)_4]$ is sp^3 , tetrahedral and diamagnetic

62. Given below are two statements :

Statement I : A mixture of $C_{12}H_{22}O_{11}$ (sugar) and NaCl can be separated by dissolving sugar in alcohol, due to differential solubility

Statement II : Rose essence from rose petals is separated by steam distillation due to its high volatility and insolubility in H_2O.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true **Statement II** is false

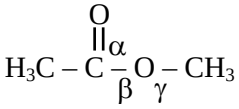
(4) **Statement I** is false **Statement II** is true

Sol. Sugar is soluble in alcohol not NaCl

So they are separated by solvent extraction (or differential extraction)

Rose essence oil is steam volatile so it is separated by steam distillation.

63. Shown below is the structure of methyl acetate with three different α, β and γ carbon-oxygen bonds.

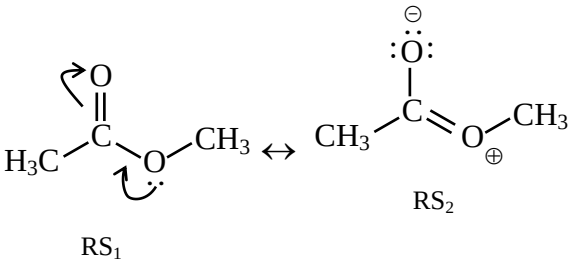


The correct order of bond length of these bonds is :

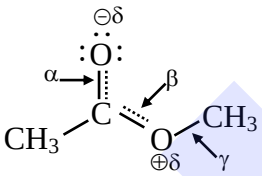
(1) α > β > γ (2) α < β < γ

(3) α = β = γ (4) α < β = γ

Sol.



RS_1 RS_2



R.Hybrid

RS1 is more stable than RS2

∴ contribution of RS1 is more than RS2 in it's R.H.

∴ Order of bond length α < β < γ

64. 'x' is the product which is obtained by the hydrolysis of prop-1-yne in the presence of mercuric sulphate under dilute acidic medium at 333 K. 'y' is the product which is obtained by the reaction of ethane nitrile with methyl magnesium bromide in dry ether followed by hydrolysis. IUPAC name of product obtained from 'x' and 'y' in the presence of barium hydroxide followed by heating is:

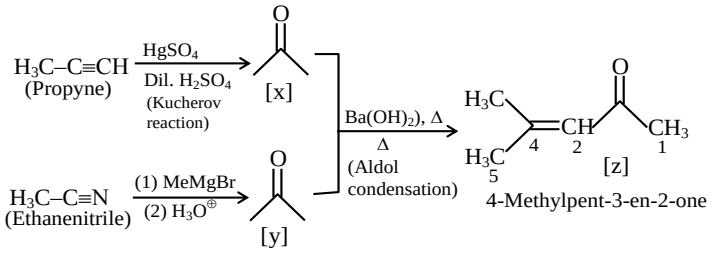
(1) 2-Methylpent-4-en-3-one

(2) 4-Methylpent-3-en-2-one

(3) 4-Methylpent-1-ene

(4) 2-Methylpent-3-one

Sol.



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65. An optically active alkyl bromide C_4H_9Br, reacts with ethanolic KOH to form major compound [A] which reacts with bromine to give compound [B]. Compound [B] reacts with ethanolic KOH and sodamide to give compound [C]. One molecule of water adds to compound [C] on warming with mercuric sulphate and dilute sulphuric acid at 333 K to form compound [D]. The functional group in compound D will be confirmed by:

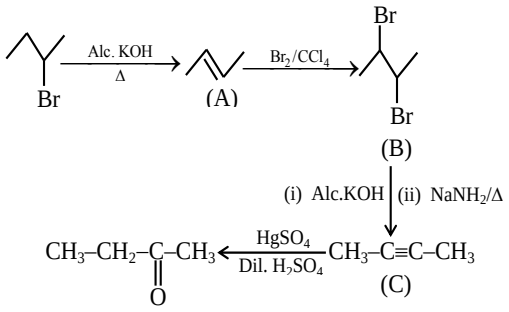
(1) Haloform test

(2) Lucas test

(3) Silver mirror test

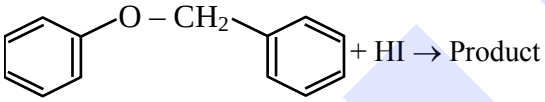
(4) Benedict test

Sol.



Final product gives +ve haloform test

66. Consider the following reaction.



Statement I : In the above reaction, product formed will be a mixture of benzyl alcohol and iodobenzene.

Statement II : In the above reaction, the – O – CH_2 – bond is cleaved to give the product.

In the light of the above statements, choose the **correct** answer from the options given below:

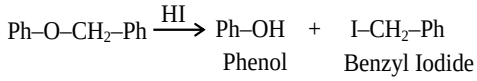
(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

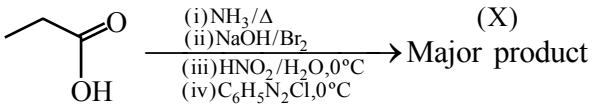
(3) **Statement I** is true but **Statement II** is false

(4) **Statement I** is false but **Statement II** is true

Sol.



67. Consider the following organic reaction sequence. Choose the final product (X) from the following (consider the major product in all intermediate reactions)



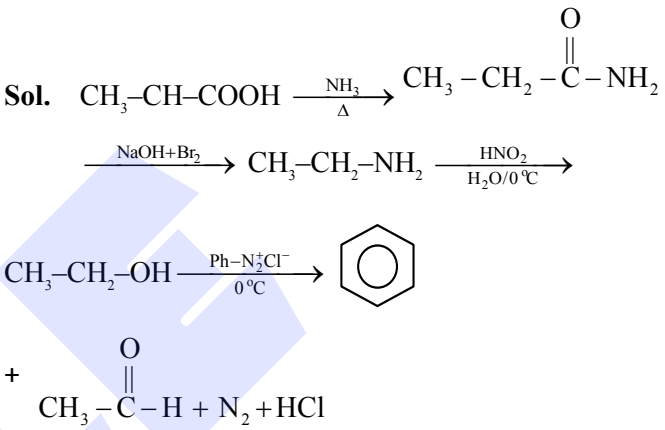
(1) Benzene

(2) Phenol

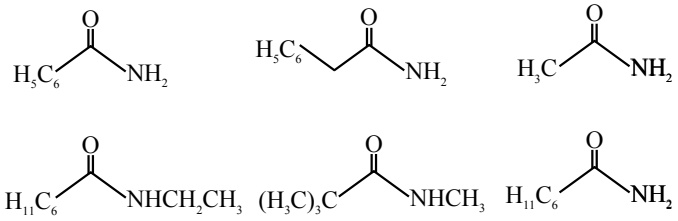
(3) Propanol

(4) Chlorobenzene

Sol.



68. The number of compounds from the following which can undergo reaction with Br_2/KOH/(alcoholic) to give respective products and these respective products can also be obtained separately by Gabriel phthalimide reaction is:



(1) 5

(2) 4

(3) 3

(4) 6

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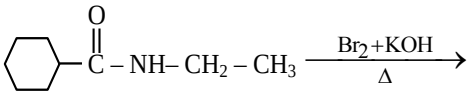
Sol. (ii), (iii), (vi)

(i) $Ph-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow[\Delta]{Br_2 + KOH} Ph-NH_2$

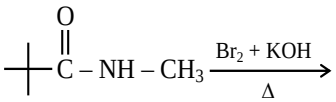
Aniline is not formed by Gabriel phtalimide synthesis.

(ii) $Ph-CH_2-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow[\Delta]{Br_2 + KOH} Ph-CH_2-NH_2$

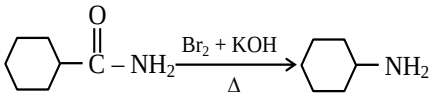
(iii) $CH_3-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow[\Delta]{Br_2 + KOH} CH_3-NH_2$

(iv) 

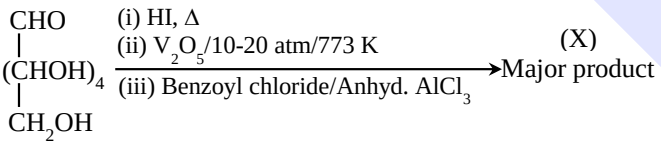
No 1° amine is form

(v) 

No 1° amine is form

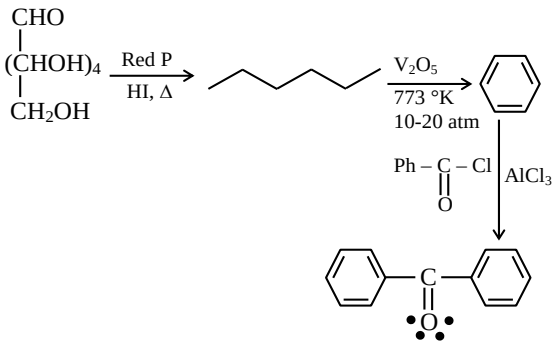
(vi) 

69. Consider the following reactions. Total number of electrons in the π bonds and lone pair of electrons in the product (X) is:



(1) 12
(2) 16
(3) 14
(4) 18

Sol.



Total number of π and lone pair electrons in final product = 18

70. Treatment of a gas 'X' with a freshly prepared ferrous sulphate solution gives a compound 'Y' as a brown ring. The compounds X and Y are.
(1) NO and $[Fe(NO)]SO_4$
(2) NO_2 and $[Fe(NO_2)]SO_4$
(3) N_2O and $[Fe(N_2O)]SO_4$
(4) N_2O_4 and $[Fe(N_2O_4)]SO_4$

Sol. $NO + FeSO_4 \xrightarrow{\text{dil. } H_2SO_4} [Fe^{I}(NO)]SO_4$

SECTION-B

71. An excess of $AgNO_3$ is added to 100 mL of a 0.05 M solution of tetraaquadichloridochromium (III) chloride. The number of moles of AgCl precipitated will be_____ $\times 10^{-3}$. (Nearest integer)

Sol. $[Cr(H_2O)_4Cl_2]Cl + AgNO_3 \rightarrow AgCl\downarrow + [Cr(H_2O)_4Cl_2]NO_3$

$\Rightarrow$ 1 mole of complex will precipitate 1 mole of AgCl.

$\Rightarrow$ Given moles of complex =
$M \times V = (0.05 \times 100 \times 10^{-3})$ moles

$\Rightarrow$ Moles of AgCl = 5×10^{-3} moles

$\Rightarrow x = 5$

72. An alkane (Y) requires 8 moles of oxygen for complete combustion and on chlorination with Cl_2/hv, (Y) gives only one monochlorinated product (Z). The total number of primary carbon atoms in (Y) is_____.

Sol. $C_nH_{2n+2} + \left(\frac{3n+1}{2}\right)O_2 \rightarrow nCO_2 + (n+1)H_2O$

$\frac{3n+1}{2} = 8$

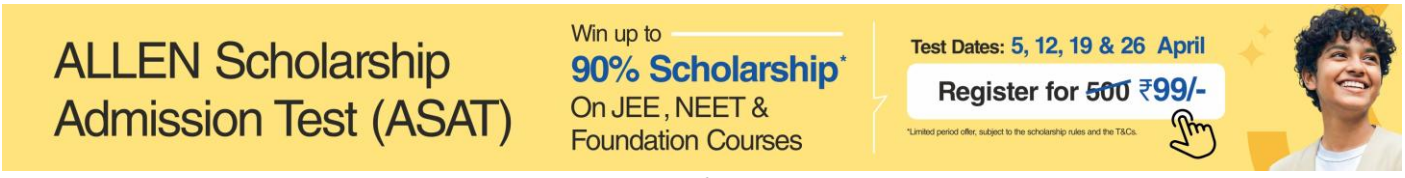
$3n + 1 = 8 \times 2$

$n = 5$ $(\therefore C_5H_{12})$

$CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3 \xrightarrow{Br_2/hv} CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-Br$

Neopentane (X) — Only one mono-bromo product

Number of primary carons = 4



73. 500 mL of 0.2 M MnO_4^- solution in basic medium when mixed with 500 mL of 1.5 M KI solution, oxidises iodide ions to liberate molecular iodine. This liberated iodine is then titrated with a standard x M thiosulphate solution in presence of starch till the end point. If 300 mL of thiosulphate was consumed, then the value of x is_____.

Sol. $3e^- + MnO_4^- + 2H_2O \rightarrow MnO_2 + 4OH^\ominus$

$2I^\ominus \rightarrow I_2 + 2e^-$

$I_2 + 2S_2O_3^{-2} \rightarrow 2I^\ominus + S_4O_6^{-2}$

gm equivalent of MnO_4^-

$= 0.2 \times \frac{500}{1000} \times 3 = 0.3$ (Limiting reagent)

gm equivalent of KI

$= 1.5 \times \frac{500}{1000} \times 1 = 0.75$

gm equivalent of MnO_4^-

= gm equivalent of hypo

$0.3 = x \times \frac{300}{1000} \times 1$

$x = 1$

74. In a closed flask at 600 K, one mole of $X_2Y_4(g)$ attains equilibrium as given below:

$X_2Y_4(g) \rightleftharpoons 2XY_2(g)$

At equilibrium, 75% $X_2Y_4(g)$ was dissociated and the total pressure is 1 atm. The magnitude of $\Delta_rG^\ominus$ (in kJ mol^{-1}) at this temperature is_____. (Nearest Integer)

(Given: R = 8.3 J mol^{-1} K^{-1}; ln 10 = 2.3, $\log 2 = 0.3, \log 3 = 0.48, \log 5 = 0.69, \log 7 = 0.84$)

Sol. $X_2Y_4(g) \rightleftharpoons 2XY_2(g)$

t=0 P^o

t=t $P^o(1-\alpha)$ $2P^o\alpha$

$P^o(1+\alpha) = 1$

$\alpha = \frac{3}{4}$; $P^o = \frac{4}{7}$

$k_P = \frac{4P_o\alpha^2}{1-\alpha^2} = \frac{36}{7}$

$\Delta G° = -RT\ell nk_P$

$|\Delta G^o| = 8.314 \times 600\ \ell n\left(\frac{36}{7}\right)$

$= 8169.5$ J mol^{-1}

$= 8.169$ kJ/mol

75. Decomposition of a hydrocarbon follows the equation $k = (5.5 \times 10^{11}\ s^{-1})\ e^{\frac{-28000K}{T}}$. The activation energy of reaction is_____ kJ mol^{-1}. (Nearest Integer)

Given : R = 8.3 JK^{-1} mol^{-1}

Sol. $K = (5.5 \times 10^{11}\ S^{-1})\ e^{\frac{-28000K}{T}}$

$K = Ae^{-E_a/RT}$

$\frac{E_a}{R} = 28000$

$E_a = \frac{28000 \times 8.3}{1000}$ kJ / mol $= 232.4$ kJ / mol

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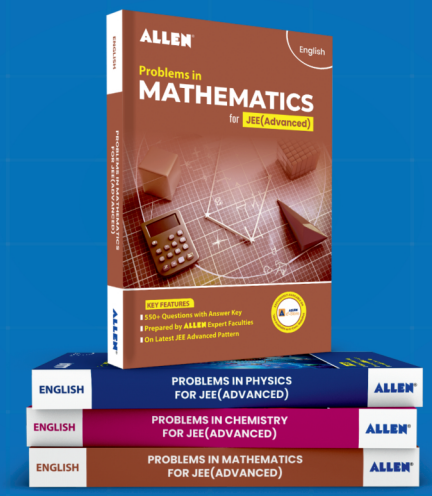


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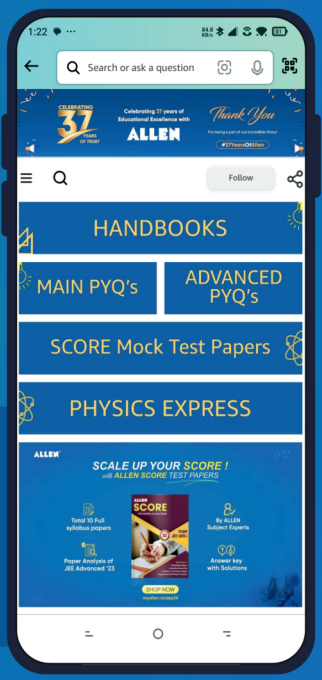
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